

Package #172

Integrate a self-expanding metal mesh sleeve around the floor lumen of the drainage catheter that flares outward to maintain a larger cross-section and resist collapse/occlusion.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'lumen occlusion' concerning multi-lumen drainage catheter. User-submitted engineering problem (session input): "Why do multi-lumen drainage catheters lose flow, and can a wider floor lumen restore drainage? (beats baseline)". [fda_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'A dual-lumen microcatheter for minimizing particle reflux during embolization: Proof-of-concept with multiphysics simulations.'; 'Role of cerebroventricular size and surgical placement in modulating catheter flow distribution.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Flared self-expanding metal stent that exerts outward radial force to keep a conduit open, a principle used in expandable pipe supports in the oil & gas industry.

Contents:

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Status: ENGINEERING_DEFINITION. Not physically validated.